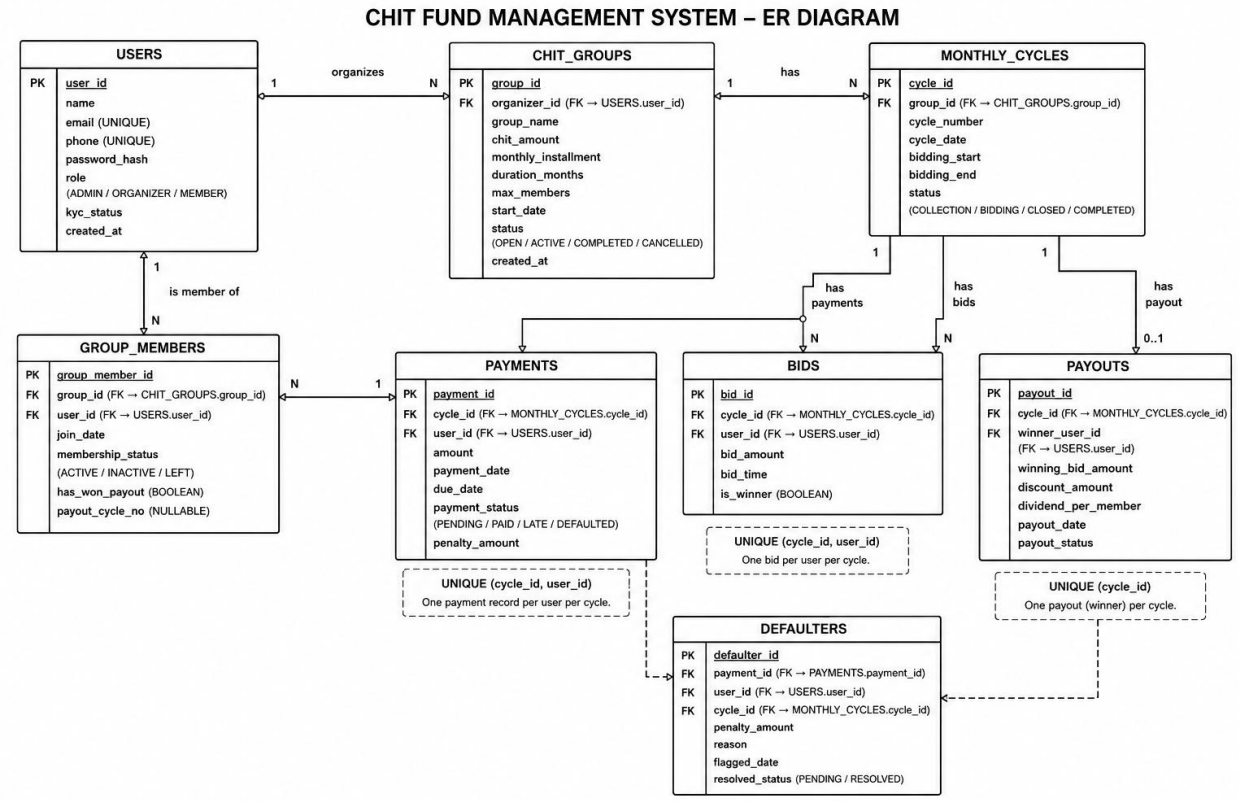
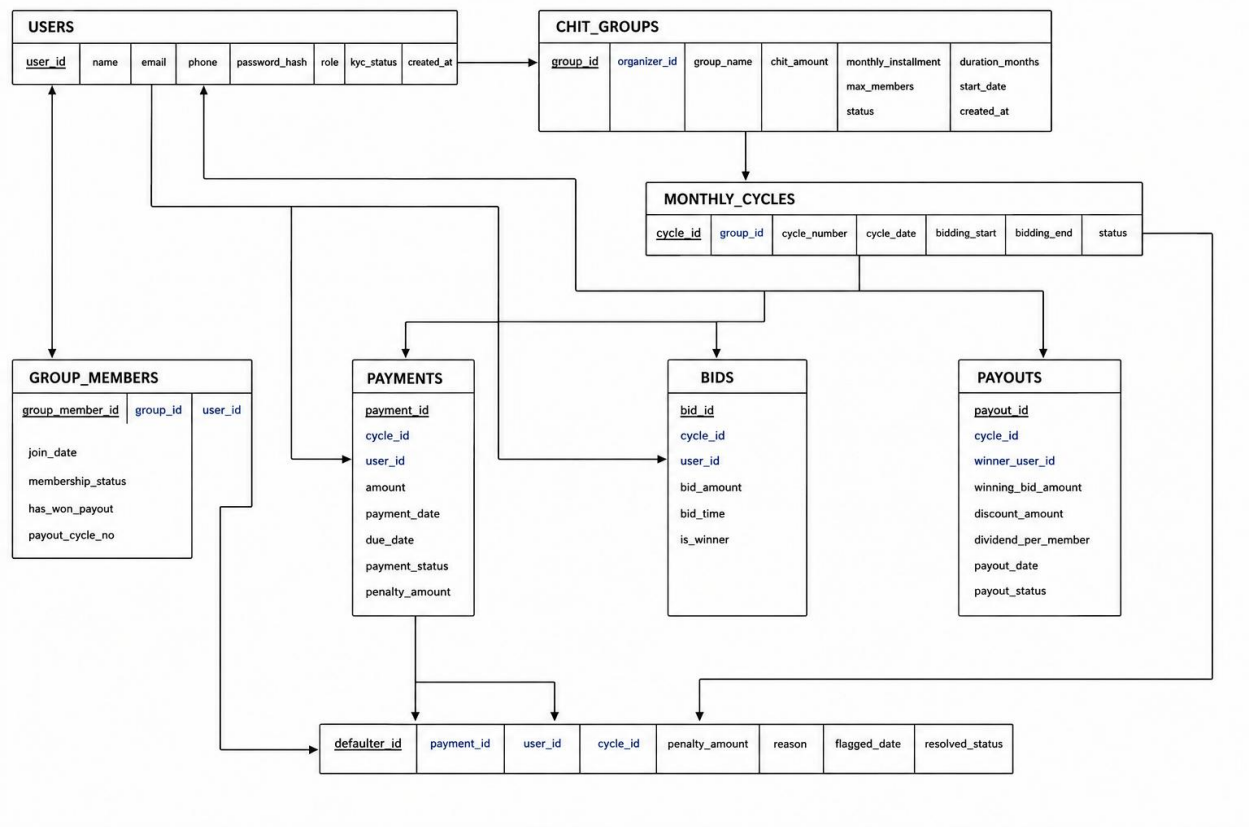
A Database-Driven Chit Fund Management and Auction System

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Section:H

ER Diagram:



Relational schema:



DDL :

```
mysql> CREATE DATABASE IF NOT EXISTS chitvault;
Query OK, 1 row affected (0.09 sec)

mysql> USE chitvault;
Database changed
```

```
mysql> CREATE TABLE Users (
    ->     user_id         INT AUTO_INCREMENT PRIMARY KEY,
    ->     name            VARCHAR(100) NOT NULL,
    ->     email           VARCHAR(100) NOT NULL UNIQUE,
    ->     phone           VARCHAR(15)  NOT NULL UNIQUE,
    ->     password_hash   VARCHAR(255) NOT NULL,
    ->     role            ENUM('organizer','member','admin') NOT NULL DEFAULT 'member',
    ->     kyc_verified    BOOLEAN DEFAULT FALSE,
    ->     created_at      TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.17 sec)
```

```
mysql> CREATE TABLE KYCDocuments (
    ->     doc_id          INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id         INT NOT NULL,
    ->     doc_type        VARCHAR(50) NOT NULL,          -- Aadhar, PAN, etc.
    ->     doc_number      VARCHAR(50) NOT NULL,
    ->     verified_status ENUM('pending','verified','rejected') DEFAULT 'pending',
    ->     uploaded_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (user_id) REFERENCES Users(user_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> CREATE TABLE ChitGroups (
    ->     group_id           INT AUTO_INCREMENT PRIMARY KEY,
    ->     group_name         VARCHAR(100) NOT NULL,
    ->     organizer_id       INT NOT NULL,
    ->     total_amount       DECIMAL(12,2) NOT NULL,
    ->     monthly_installment DECIMAL(10,2) NOT NULL,
    ->     duration_months    INT NOT NULL,
    ->     max_members        INT NOT NULL,
    ->     start_date         DATE NOT NULL,
    ->     status             ENUM('upcoming','active','completed','cancelled') DEFAULT 'upcoming',
    ->     created_at         TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (organizer_id) REFERENCES Users(user_id),
    ->     CHECK (total_amount = monthly_installment * duration_months)
    -> );
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> CREATE TABLE GroupMembers (
    ->     membership_id   INT AUTO_INCREMENT PRIMARY KEY,
    ->     group_id        INT NOT NULL,
    ->     user_id         INT NOT NULL,
    ->     joined_date     DATE DEFAULT (CURRENT_DATE),
    ->     has_won_payout  BOOLEAN DEFAULT FALSE,
    ->     slot_status     ENUM('active','defaulted','exited') DEFAULT 'active',
    ->     FOREIGN KEY (group_id) REFERENCES ChitGroups(group_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
```

```
mysql> CREATE TABLE MonthlyCycles (
    ->     cycle_id           INT AUTO_INCREMENT PRIMARY KEY,
    ->     group_id           INT NOT NULL,
    ->     month_number       INT NOT NULL,
    ->     cycle_date         DATE NOT NULL,
    ->     status             ENUM('collection','bidding','closed') DEFAULT 'collection',
    ->     total_pool_amount  DECIMAL(12,2),
    ->     winning_bid_amount DECIMAL(12,2),
    ->     FOREIGN KEY (group_id) REFERENCES ChitGroups(group_id) ON DELETE CASCADE,
    ->     UNIQUE (group_id, month_number)     -- one cycle per month per group
    -> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> CREATE TABLE Payments (
    ->     payment_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     cycle_id        INT NOT NULL,
    ->     user_id         INT NOT NULL,
    ->     amount_paid     DECIMAL(10,2) NOT NULL,
    ->     payment_date    DATE DEFAULT (CURRENT_DATE),
    ->     status          ENUM('paid','late','defaulted') DEFAULT 'paid',
    ->     penalty_amount  DECIMAL(10,2) DEFAULT 0,
    ->     FOREIGN KEY (cycle_id) REFERENCES MonthlyCycles(cycle_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
    ->     UNIQUE (cycle_id, user_id)          -- one payment per member per cycle
    -> );
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> CREATE TABLE Bids (
    ->     bid_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     cycle_id    INT NOT NULL,
    ->     user_id     INT NOT NULL,
    ->     bid_amount  DECIMAL(10,2) NOT NULL,
    ->     bid_time    TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     is_winner   BOOLEAN DEFAULT FALSE,
    ->     FOREIGN KEY (cycle_id) REFERENCES MonthlyCycles(cycle_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
    ->     UNIQUE (cycle_id, user_id)          -- one bid per member per cycle
```

```
mysql> CREATE TABLE Payouts (
    ->     payout_id           INT AUTO_INCREMENT PRIMARY KEY,
    ->     cycle_id            INT NOT NULL UNIQUE,     -- exactly one payout per cycle
    ->     winner_id           INT NOT NULL,
    ->     amount_disbursed    DECIMAL(12,2) NOT NULL,
    ->     dividend_per_member DECIMAL(10,2),
    ->     disbursed_date      DATE DEFAULT (CURRENT_DATE),
    ->     FOREIGN KEY (cycle_id)  REFERENCES MonthlyCycles(cycle_id) ON DELETE CASCADE,
    ->     FOREIGN KEY (winner_id) REFERENCES Users(user_id)
    -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> CREATE TABLE Defaulters (
    ->     default_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id         INT NOT NULL,
    ->     group_id        INT NOT NULL,
    ->     cycle_id        INT NOT NULL,
    ->     amount_due      DECIMAL(10,2) NOT NULL,
    ->     penalty_applied DECIMAL(10,2) DEFAULT 0,
    ->     resolved_status ENUM('pending','resolved') DEFAULT 'pending',
    ->     FOREIGN KEY (user_id)  REFERENCES Users(user_id),
    ->     FOREIGN KEY (group_id) REFERENCES ChitGroups(group_id),
    ->     FOREIGN KEY (cycle_id) REFERENCES MonthlyCycles(cycle_id)
    -> );
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> CREATE TABLE Notifications (
    ->     notif_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     user_id     INT NOT NULL,
    ->     message     VARCHAR(255) NOT NULL,
    ->     type        VARCHAR(50),
    ->     sent_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     read_status BOOLEAN DEFAULT FALSE,
    ->     FOREIGN KEY (user_id) REFERENCES Users(user_id) ON DELETE CASCADE
    -> );
```

```
mysql> DELIMITER //
mysql>
mysql> -- Prevent a member who already won a payout in this group from bidding again
mysql> CREATE TRIGGER trg_prevent_repeat_bid
    -> BEFORE INSERT ON Bids
    -> FOR EACH ROW
    -> BEGIN
    ->     DECLARE already_won BOOLEAN;
    ->     SELECT gm.has_won_payout INTO already_won
    ->     FROM GroupMembers gm
    ->     JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    ->     WHERE mc.cycle_id = NEW.cycle_id AND gm.user_id = NEW.user_id;
    ->
    ->     IF already_won = TRUE THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Member has already won a payout in this group and cannot bid again';
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> -- Auto-flag a defaulter when a payment is marked 'defaulted'
mysql> CREATE TRIGGER trg_flag_defaulter
    -> AFTER UPDATE ON Payments
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.status = 'defaulted' AND OLD.status <> 'defaulted' THEN
    ->         INSERT INTO Defaulters (user_id, group_id, cycle_id, amount_due, penalty_applied)
    ->         SELECT NEW.user_id, mc.group_id, NEW.cycle_id, NEW.amount_paid, NEW.penalty_amount
    ->         FROM MonthlyCycles mc WHERE mc.cycle_id = NEW.cycle_id;
    ->     END IF;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> -- Lock the winner's flag once a payout is created for a cycle
mysql> CREATE TRIGGER trg_lock_winner
    -> AFTER INSERT ON Payouts
    -> FOR EACH ROW
    -> BEGIN
    ->     UPDATE GroupMembers gm
    ->     JOIN MonthlyCycles mc ON mc.group_id = gm.group_id
    ->     SET gm.has_won_payout = TRUE
    ->     WHERE mc.cycle_id = NEW.cycle_id AND gm.user_id = NEW.winner_id;
    -> END//
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> SHOW TABLES;
+--------------------+
| Tables_in_chitvault |
+--------------------+
| bids               |
| chitgroups         |
| defaulters         |
| groupmembers       |
| kycdocuments       |
| monthlycycles      |
| notifications      |
| payments           |
| payouts            |
| users              |
+--------------------+
10 rows in set (0.04 sec)
```

```
mysql> DESCRIBE ChitGroups;
+---------------------+-----------------------------------------------------+------+-----+-------------------+-------------------+
| Field               | Type                                                | Null | Key | Default           | Extra             |
+---------------------+-----------------------------------------------------+------+-----+-------------------+-------------------+
| group_id            | int                                                 | NO   | PRI | NULL              | auto_increment    |
| group_name          | varchar(100)                                        | NO   |     | NULL              |                   |
| organizer_id        | int                                                 | NO   | MUL | NULL              |                   |
| total_amount        | decimal(12,2)                                       | NO   |     | NULL              |                   |
| monthly_installment | decimal(10,2)                                       | NO   |     | NULL              |                   |
| duration_months     | int                                                 | NO   |     | NULL              |                   |
| max_members         | int                                                 | NO   |     | NULL              |                   |
| start_date          | date                                                | NO   |     | NULL              |                   |
| status              | enum('upcoming','active','completed','cancelled') | YES  |     | upcoming          |                   |
| created_at          | timestamp                                           | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------------+-----------------------------------------------------+------+-----+-------------------+-------------------+
10 rows in set (0.03 sec)
```

```
mysql> DESCRIBE Payouts;
+---------------------+---------------+------+-----+------------+-------------------+
| Field               | Type          | Null | Key | Default    | Extra             |
+---------------------+---------------+------+-----+------------+-------------------+
| payout_id           | int           | NO   | PRI | NULL       | auto_increment    |
| cycle_id            | int           | NO   | UNI | NULL       |                   |
| winner_id           | int           | NO   | MUL | NULL       |                   |
| amount_disbursed    | decimal(12,2) | NO   |     | NULL       |                   |
| dividend_per_member | decimal(10,2) | YES  |     | NULL       |                   |
| disbursed_date      | date          | YES  |     | curdate()  | DEFAULT_GENERATED |
+---------------------+---------------+------+-----+------------+-------------------+
6 rows in set (0.01 sec)
```

```
mysql> SHOW TRIGGERS;
+----------------------+--------+----------+-----------------------------------------------------
```